

Introduction to Spectral Methods

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Outline

- 1 **Basic principles**
 - Problem formulation
 - Various numerical methods
 - Various spectral methods
 - How to choose trial functions
- 2 **Two illustrative examples of spectral methods**
 - A Fourier Galerkin method for the wave equation
 - A Chebyshev collocation method for the heat equation
- 3 **Summary**

Solving a PDE numerically

Consider the PDE with boundary condition

$$\begin{aligned}Lu &= f, \text{ in } \Omega \\ u &= g, \text{ on } \partial\Omega.\end{aligned}$$

Question: How to solve the above PDE *numerically*?
Approximate the unknown $u(x, t)$ by a sum of “basis functions”:

$$u(x, t) \approx u^N(x, t) = \sum_{n=0}^N a_n(t) \phi_n(x)$$

and use some strategy to minimize the *Residual* $Lu^N - f$.

Trial functions and test functions

Search for solution u^N in a finite-dimensional sub-space H_N of some Hilbert space H .

Trial Functions: basis of H_N : $(\phi_0, \dots, \phi_N)$

$$u^N = \sum_{k=0}^N a_k \phi_k$$

Test Functions: family of functions $(\psi_0, \dots, \psi_N)$

$$\forall n \in 0, \dots, N, (\psi_n, R) = 0$$

Classification according to trial functions

- **Finite difference:** trial functions: overlapping **local** polynomials of low order
- **Finite element:** trial functions: **local** smooth functions, nearly orthogonal
- **Spectral methods:** trial functions: **global** smooth functions, nearly orthogonal

Classification according to **test functions**

- **Galerkin**: $\psi_n = \phi_n$, ϕ_n satisfy some or all of the boundary conditions.
- **Collocation**: $\psi_n = \delta(x - x_n)$, x_n are *collocation* points.
- **Tau**: $\psi_n = \phi_n$, but ϕ_n do not satisfy the boundary conditions.

What sets of "trial functions" will work?

It is obvious that we would like our trial function sets to have a number of properties:

- easy to compute
- rapid convergence
- completeness

Outline of the second part

- A Fourier Galerkin method for the wave equation:
 - trial function
 - test function
 - weak formulation
 - accuracy
 - comparison with FD
- A Chebyshev collocation method for the heat equation:
..., comparison with Galerkin method, ...

Wave equation

Many evolution equations can be written as

$$\frac{\partial u}{\partial t} = M(u).$$

Consider the domain $(0, 2\pi)$ with **periodic** boundary conditions.
The approximate solution u^N is represented as

$$u^N(x, t) = \sum_{-N/2}^{N/2} a_k(t) \phi_k(x).$$

Weak formulation

In general,

$$\frac{\partial u^N}{\partial t} \neq M(u^N).$$

The approximation is obtained by selecting a set of test functions ψ_k and requiring that

$$\int_0^{2\pi} \left[\frac{\partial u^N}{\partial t} - M(u^N) \right] \psi_k(x) dx = 0, \quad (1)$$

for $k = -N/2, \dots, N/2$.

Spectral method using trigonometric polynomials

Trigonometric polynomials:

$$\begin{aligned}\phi_k(x) &= e^{ikx}, \\ \psi_k(x) &= \frac{1}{2\pi} e^{-ikx}.\end{aligned}$$

If this were merely an approximation problem, then $u^N(x, t)$ would be the truncated Fourier series of the known function $u(x, t)$ with

$$a_k(t) = \int_0^{2\pi} u(x, t) \psi_k(x) dx.$$

However, for the PDE, $u(x, t)$ is not known; the approximation is determined by (1).

A Fourier Galerkin method for the wave equation

How does the scheme work (1) ?

For the linear hyperbolic problem

$$\frac{\partial u}{\partial t} - \frac{\partial u}{\partial x} = 0,$$

i.e., for

$$M(u) = \frac{\partial u}{\partial x},$$

condition (1) becomes

$$\frac{1}{2\pi} \int_0^{2\pi} \left[\left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x} \right) \sum_{-N/2}^{N/2} a_l(t) e^{ilx} \right] e^{-ikx} dx = 0,$$

How does the scheme work (2)?

The analytical (spatial) differentiation of the trial functions and the analytical integration of that expression produce the dynamical equations:

$$\frac{da_k}{dt} - ika_k = 0, k = -N/2, \dots, N/2.$$

The initial conditions are:

$$a_k(0) = \int_0^{2\pi} u(x, 0) \psi_k(x) dx.$$

The accuracy of the Fourier Galerkin method (1)

Use the initial condition

$$u(x, 0) = \sin(\pi \cos(x))$$

to illustrate the accuracy of the Fourier Galerkin method for the above hyperbolic equation. The exact solution,

$$u(x, t) = \sin(\pi \cos(x + t)),$$

has the Fourier expansion

$$u(x, t) = \sum_{k=-\infty}^{\infty} a_k(t) e^{ikx},$$

The accuracy of the Fourier Galerkin method (2)

The Fourier coefficients are

$$a_k(t) = \sin\left(\frac{k\pi}{2}\right) J_k(\pi) e^{ikt}$$

and $J_k(t)$ is the Bessel function of order k .

The **asymptotic properties** of the Bessel functions imply that

$$k^p a_k(t) \rightarrow 0 \quad \text{as } k \rightarrow \infty$$

for all positive integers p . Thus, the truncated Fourier series,

$$u^N(x, t) = \sum_{-N/2}^{N/2} a_k(t) e^{ikx}$$

converges faster than any finite power of $1/N$.

A Fourier Galerkin method for the wave equation

Comparison with finite difference method

An illustrative of the superior accuracy from the spectral method for this problem is given in the following figure.

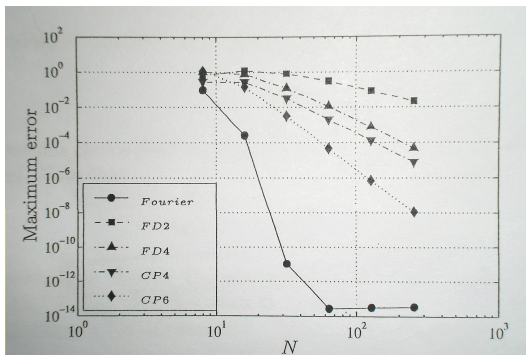


Figure: Maximum errors for the linear hyperbolic problem at $t = 2\pi$ for Fourier Galerkin and several finite difference schemes

A Chebyshev collocation method for the heat equation

Chebyshev polynomials for the heat equation

Chebyshev polynomials:

$$T_k(x) = \cos(k \cos^{-1} x), \quad \text{for } k = 0, 1, \dots$$

The first few Chebyshev polynomials are

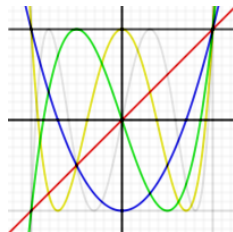
$$T_0(x) = 1$$

$$T_1(x) = x$$

$$T_2(x) = 2x^2 - 1$$

...

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$



A Chebyshev collocation method for the heat equation

The approximate solution (1)

Given the heat equation

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0, \text{ i.e., } M(u) = \frac{\partial^2 u}{\partial x^2}$$

on $(-1, 1)$ with homogeneous Dirichlet boundary conditions,

$$u(-1, t) = 0, \quad u(1, t) = 0.$$

Choosing the **trial functions**

$$\phi_k(x) = T_k(x), \quad k = 0, 1, \dots, N,$$

the approximate solution has the representation

$$u^N(x, t) = \sum_{k=0}^N a_k(t) \phi_k(x).$$

A Chebyshev collocation method for the heat equation

The approximate solution (2)

In the collocation approach the above PDE must be satisfied exactly by the approximate solution at *collocation* points x_j in the domain of $(-1, 1)$:

$$\frac{\partial u^N}{\partial t} - M(u^N)|_{x=x_j} = 0, \quad j = 1, \dots, N-1. \quad (2)$$

$$u^N(-1, t) = 0, \quad u^N(1, t) = 0.$$

$$u^N(x_k, 0) = u(x_k, 0), \quad k = 0, \dots, N.$$

A Chebyshev collocation method for the heat equation

Choice of collocation points

A convenient choice for the collocation points x_j is

$$x_j = \cos\left(\frac{\pi j}{N}\right).$$

Note that

$$\phi_k(x_j) = \cos\left(\frac{\pi jk}{N}\right).$$

We can apply *Fast Fourier Transform (FFT)* to evaluate $M(u^N)|_{x=x_j}$.

A Chebyshev collocation method for the heat equation

How the Chebyshev collocation approach works (1)

For the particular initial condition

$$u(x, 0) = \sin \pi x,$$

the exact solution is

$$u(x, t) = e^{-\pi^2 t} \sin \pi x.$$

It has the infinite Chebyshev expansion

$$u(x, t) = \sum_{k=0}^{\infty} b_k(t) T_k(x),$$

A Chebyshev collocation method for the heat equation

How the Chebyshev collocation approach works (2)

... where

$$b_k(t) = \frac{2}{c_k} \sin\left(\frac{k\pi}{2}\right) J_k(\pi) e^{-\pi^2 t}$$

with

$$c_k = \begin{cases} 2, & k = 0, \\ 1, & k \geq 1 \end{cases}$$

Since $J_k(\pi)$ is decaying rapidly, the truncated series converges at an **exponential** rate. A well-designed collocation method will do the same.

A Chebyshev collocation method for the heat equation

How the Chebyshev collocation approach works (3)

A collocation method is implemented in terms of the **nodal values** $u_j(t) = u^N(x_j, t)$ and we have the expansion

$$u^N(x, t) = \sum_{j=0}^N u_j(t) \phi_j(x),$$

and now ϕ_j denote the **characteristic Lagrange polynomials** with the property $\phi_j(x_i) = \delta_{ij}$ for $0 \leq i, j \leq N$. The expansion coefficients are given by

$$a_k(t) = \frac{2}{N\bar{c}_k} \sum_{l=0}^N \bar{c}_l^{-1} u_l(t) \cos \frac{\pi l k}{N}, \quad k = 0, 1, \dots, N,$$

where

$$\bar{c}_k = \begin{cases} 2, & k = 0 \text{ or } N \\ 1, & 1 \leq k \leq N-1 \end{cases}$$

A Chebyshev collocation method for the heat equation

How the Chebyshev collocation approach works (4)

The exact derivative of $u^N(x, t)$ becomes

$$\frac{\partial^2 u^N}{\partial x^2}(t) = \sum_{k=0}^N a_k^{(2)}(t) T_k(x),$$

where

$$\begin{aligned} a_{N+1}^{(1)}(t) &= 0, \quad a_N^{(1)}(t) = 0, \\ \bar{c}_k a_k^{(1)}(t) &= a_{k+2}^{(1)}(t) + 2(k+1)a_{k+1}^{(1)}(t), \quad k = N-1, \dots, 0, \end{aligned}$$

and

$$\begin{aligned} a_{N+1}^{(2)}(t) &= 0, \quad a_N^{(2)}(t) = 0, \\ \bar{c}_k a_k^{(2)}(t) &= a_{k+2}^{(2)}(t) + 2(k+1)a_{k+1}^{(2)}(t), \quad k = N-1, \dots, 0. \end{aligned}$$

A Chebyshev collocation method for the heat equation

How the Chebyshev collocation approach works (5)

The coefficients $a_k^{(2)}$ depend linearly on the nodal values u_l ; thus, there exists a matrix D_N^2 such that

$$\frac{\partial^2 u^N}{\partial x^2}(t)|_{x=x_j} = \sum_{k=0}^N a_k^{(2)}(t) \cos \frac{\pi j k}{N} = \sum_{l=0}^N (D_N^2)_{jl} u_l(t).$$

Substituting the above expression into (2), we obtain a system of ODE for the nodal unknowns:

$$\frac{du_j}{dt}(t) = \sum_{l=0}^N (D_N^2)_{jl} u_l(t), \quad j = 1, \dots, N-1.$$

Supplemented by the initial conditions, the ODE system for the nodal values of solution is readily integrated in time.

A Chebyshev collocation method for the heat equation

Comparison with Finite difference method

An illustrative of the superior accuracy from the spectral method for this problem is given in the following figure.

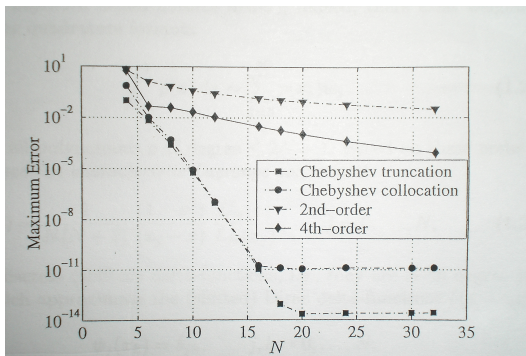


Figure: Maximum errors for the heat equation problem at $t = 1$ for Chebyshev collocation and several finite difference schemes

Pros and Cons of spectral methods

Spectral methods have many advantages over FD and FE methods:

- high accuracy
- efficiency
- exponential convergency/spectral convergency

However, spectral methods also suffers drawbacks in the following folds:

- coding: more difficult to code than FD
- cost: costly per degree of freedom than FD
- geometry: for complicated domains, heavy loss of accuracy

Thanks for your attention!